



**Birla Vishvakarma Mahavidyalaya Engineering College**

**The Robotics Society, India Student chapter**

Report on

# Expert Lecture on "FROM BLUEPRINT TO BOT: A BEGINNER'S JOURNEY INTO ROBOTICS"

DATE AND DAY:- 6th August, 2025, WEDNESDAY

**FROM BLUEPRINT TO BOT:  
A BEGINNERS JOURNEY INTO ROBOTICS**

Discover the foundations of robotics with expert mentor. Explore the process of designing, building, and presenting, and unlock the skills to grow and innovate

**6TH AUGUST 2025**  
**4:00 PM ONWARDS**  
**D-101**

**PROF. (DR.) ASHISH M. THAKKAR**  
EXPERT FACULTY

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The TRS Student Chapter of Birla Vishvakarma Mahavidyalaya (BVM), under the Charutar Vidya Mandal (CVM), successfully organized an informative expert lecture on **“From Blueprint to Bot: A Beginner’s Journey into Robotics”** on 6th August 2025 at BVM, Vallabh Vidyanagar.

The session was delivered by **Prof. (Dr.) Ashish M. Thakkar**, who shared his valuable knowledge and experience in the field of robotics. The lecture mainly focused on introducing students to the fundamental building blocks of robotics and guiding them through the systematic journey of transforming a conceptual blueprint into a fully functional robotic system. His session was highly interactive and beginner-friendly, helping students understand complex concepts in a simplified and practical manner.

Prof. Thakkar guided participants through the core components of robotics such as sensors, actuators, and motors, explaining their working principles and importance in robotic applications. He elaborated on how sensors act as the sensing elements of a robot, allowing it to interact with its environment by detecting parameters such as distance, light, temperature, and obstacles. The session also highlighted the role of actuators as mechanical components responsible for converting electrical energy into physical movement, enabling robots to perform required actions efficiently.

A detailed explanation of different types of motors used in robotics, including DC motors, servo motors, and stepper motors, was provided during the lecture. Prof. Thakkar discussed the selection criteria of motors based on torque, speed, accuracy, and application requirements. He also demonstrated how the proper integration of sensors, actuators, and motors with controllers forms the backbone of any successful robotic system. The explanation helped students understand the coordination between hardware components and control mechanisms in real-time robotic operations.

The lecture also covered the systematic design and development process of robotics projects. Students were introduced to the stages of conceptual planning, design modeling, component selection, testing, troubleshooting, and final presentation. Prof. Thakkar emphasized that robotics is a multidisciplinary domain requiring knowledge of mechanical design, electronics, programming, and teamwork.

Apart from technical learning, the session also focused on an important aspect of robotics competitions, which is structured report making and documentation. Prof. Thakkar provided detailed guidance on preparing technical reports for national-level competitions such as **Robofest** and **Robocon**. He explained the importance of presenting projects in a professional and well-organized manner, including sections such as abstract, problem statement, methodology, design explanation, results, and conclusion. He also emphasized the importance of diagrams, block representations, innovation justification, and proper formatting in technical reports. This guidance helped students understand how documentation plays a vital role in competition evaluation and project presentation.



An interactive question and answer session further enhanced the learning experience. Students actively participated by asking questions related to project development, competition preparation, skill requirements, and career opportunities in robotics. Prof. Thakkar addressed the queries with practical examples and motivational insights, encouraging students to actively explore robotics and innovation-based learning.

The event was graced by the presence of respected faculty members including Prof. (Dr.) Vinay Patel, Principal of BVM and Advisor of the TRS Student Chapter, along with Prof. (Dr.) Haresh Patoliya, Prof. (Dr.) Mehfuza Holia, Prof. Jayesh Koisha, Prof. (Dr.) M. M. Solanki, and Prof. (Dr.) Jagruti Shah, whose valuable presence and constant encouragement contributed to the success of the session.

The event was efficiently coordinated by Prof. (Dr.) Mehfuza Holia with the enthusiastic support of the student organizing team including Utsav Maradia, Ruchit Soni, Prachi Bachani, Kishan Mori, and Rushik Javiya. The dedicated efforts of the organizing team ensured smooth execution of the program, proper technical arrangements, and effective management of participants.

The TRS Student Chapter and organizing team expressed their sincere gratitude to Prof. (Dr.) Vinay Patel for his continuous guidance and support. Special appreciation was extended to the BVM-CVM Management for encouraging technical and innovation-based activities that provide students with exposure to emerging technological domains.

The session was highly appreciated by the participants for its clarity, practical insights, and motivational approach. Many students expressed increased interest in robotics project development and participation in competitions such as Robofest and Robocon. The lecture successfully inspired students to explore hands-on learning and develop innovative solutions through robotics.

The expert lecture proved to be a knowledge-enriching and inspiring experience for all attendees, strengthening their understanding of robotics fundamentals and motivating them to pursue technical excellence and innovation in the field of engineering.